

Modeling a Single Rotary Impact

Synthesized coefficients, an unknown hammer component, and COP

Hob Nilre & Bo C. Herlin

2026-10-05

Abstract

A rotary hammer strikes an anvil against a tightened joint. We synthesize third- and fourth-derivative coefficients and assign their torque to an unknown component inside the hammer. Signed torque–speed integrals give the component’s supplied or absorbed work and apparent coefficients of performance. A separate order comparison shows how a fourth-order scalar equation recovers both modes of an ordinary two-inertia collision. In calculated impacts with 28.8 J of incoming kinetic energy, the ordinary linear model delivers 2.43 J net to the anvil; a stated fourth-order hammer model delivers 2.91 J. Doubling its fourth-order weight gives 35.39 J and an apparent anvil COP of 1.229, accompanied by a growing engaged mode. Rebound, joint work and contact release complete the ordinary energy account; the unknown component’s remaining account is a question for measurement.

PDF created: 2026-10-10 16:36:46 UTC

Latest on GitHub: <https://github.com/hobnilre/physics-impact-driver-model>

1 One blow

The interval of interest begins when a rotating hammer meets an anvil and ends when they separate. The tightened fastener undergoes no gross turning, but the anvil, socket and joint can twist. That motion determines how much work reaches the joint and how much returns to the hammer.

We add third- and fourth-derivative torque terms constructed using *ODE Coefficient Synthesis* (Nilre and Herlin, 2026a). Following *Energy Ledgers for Forced Harmonic ODEs* (Nilre and Herlin, 2026b), we assume that these terms belong to an unknown component inside the hammer. Its signed power determines whether it supplies or absorbs work. Its physical identity, capacity and preparation remain open parts of the hypothesis.

Let θ_h, θ_a be hammer and anvil angles measured from engagement, $\omega_h = \dot{\theta}_h$, $\omega_a = \dot{\theta}_a$, and $\delta = \theta_h - \theta_a$ the contact deformation. The resolved inertias are $J_h, J_a > 0$. Positive contact torque τ_c opposes forward hammer motion and drives the anvil. During contact,

$$\begin{aligned} J_h \ddot{\theta}_h + \tau_c + \tau_X &= \tau_d, \\ J_a \ddot{\theta}_a &= \tau_c - \tau_j, \\ \tau_j &= k_j \theta_a + c_j \omega_a, \\ \tau_X &= B_3 \theta_h^{(3)} + B_4 \theta_h^{(4)}. \end{aligned} \tag{1}$$

Here τ_d is externally applied hammer torque. The socket and tightened joint are represented together by $k_j, c_j > 0$, with the joint reference fixed. The unknown torque acts at the hammer coordinate, so its power uses ω_h . Figure 1 locates these accounts.

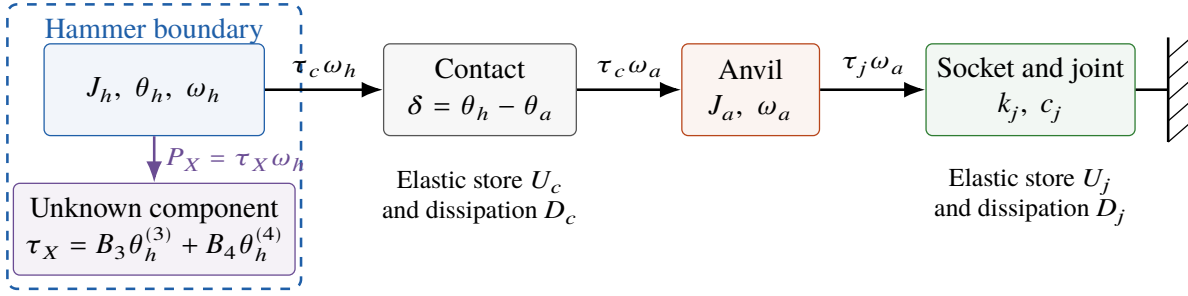


Figure 1. Schematic of the single-impact model. Contact transfers work at two different angular velocities. The unknown component is inside the hammer boundary; positive P_X enters its account. The joint reference is fixed.

For the reference comparison, $\tau_d = 0$, both angles start at zero, $\omega_h(0) = 600 \text{ rad s}^{-1}$, and $\omega_a(0) = 0$. Elastic stores are initially zero in this incremental model. The motor's earlier run-up is represented by the incoming hammer kinetic energy.

2 Coefficients and contact

For a positive rotational reference triple (J_*, c_*, k_*) , the synthesis family is

$$A_{r,n} = J_*^{n-1+r} c_*^{2-n-2r} k_*^r, \quad r \in \mathbb{Z}, \quad n \geq 0. \quad (2)$$

The derivative index is n ; the integer r selects a dimensionally admissible monomial. The companion's staircase $r_n = 1 - \lfloor n/2 \rfloor$ gives $A_0 = k_*$, $A_1 = c_*$, $A_2 = J_*$, $A_3 = J_* c_*/k_*$ and $A_4 = J_*^2/k_*$ (Nilre and Herlin, 2026a, Sections 1–4). We set

$$B_3 = w_3 \frac{J_* c_*}{k_*}, \quad B_4 = w_4 \frac{J_*^2}{k_*}. \quad (3)$$

The dimensionless weights specify the proposed model. In the examples, $(J_*, c_*, k_*) = (J_h, c_c, k_c)$ sets reference scales; the resolved hammer inertia still appears only once in (1).

The ordinary linear contact is

$$\tau_c = k_c \delta + c_c \dot{\delta}. \quad (4)$$

Switched spring–damper contact is used in impact-wrench modeling (ter Braack and Margolis, 2026). We use it until its first positive-to-negative torque zero, then set $\tau_c = 0$. This avoids extending the contact into tension. The residual spring store at release is accounted for below.

A nonlinear comparison uses the deformation-dependent damping form of Hunt and Crossley (1975):

$$\tau_c = H \delta^{3/2} (1 + \alpha \dot{\delta}), \quad \delta \geq 0. \quad (5)$$

This is a local rotary approximation to a Hertz-type contact at a constant lever arm R : displacement $R\delta$ and torque RF give $H = KR^{5/2}$ from a normal stiffness coefficient K . It represents that assumed contact geometry. Choose $H = k_c/\sqrt{\delta_r}$ and $\alpha = c_c/(k_c \delta_r)$, with $\delta_r = 0.040 \text{ rad}$. Elastic torque and the damping slope then match (4) at δ_r . Separation is the first returning zero of δ ; $1 + \alpha \dot{\delta}$ stays positive in all reported runs.

The ordinary comparisons set $B_3 = B_4 = 0$. The proposed third-order hammer sets $(w_3, w_4) = (1/2, 0)$, and the proposed fourth-order hammer uses $(1/2, 1/10)$, both with (4). A coefficient sensitivity doubles w_4 to $1/5$. These are declared illustrative weights, with no fit to measurements or target COP. The coupled third- and fourth-order models have five and six motion states respectively.

Extra order requires extra initial data. For the linear-contact examples,

$$\ddot{\theta}_h(0) = -\frac{c_c \omega_h(0)}{J_h}, \quad \theta_h^{(3)}(0) = 0 \quad \text{for fourth order.} \quad (6)$$

These choices start the unknown torque at zero; the governing equation sets the next derivative. At separation, body angles and velocities remain continuous, and the contact-activated higher-order law ends. Endpoint derivatives in its work integral are taken immediately before release. The unknown component's post-contact state belongs to its remaining account.

The added terms have both coefficients and preparation conditions. Specifying incoming speed alone does not specify the higher-order blow.

3 What fourth order recovers

Before interpreting an unknown contribution, it is useful to establish why fourth order appears in a rotary collision at all. Set $\tau_X = \tau_d = 0$ and use the linear contact. Eliminating the anvil coordinate from the two ordinary equations gives the exact engaged equation

$$\begin{aligned} J_h J_a \theta_h^{(4)} + [J_h(c_c + c_j) + J_a c_c] \theta_h^{(3)} \\ + [J_h(k_c + k_j) + J_a k_c + c_c c_j] \ddot{\theta}_h \\ + (c_c k_j + c_j k_c) \dot{\theta}_h + k_c k_j \theta_h = 0. \end{aligned} \quad (7)$$

Dividing (7) by k_* gives torque-equation units; its coefficients can then be expressed as weighted members of (2). Its extra initial data are fixed by the eliminated anvil state. This scalar representation and the proposed hammer component in (1) have different physical assignments: the latter adds a new torque to the resolved two-body equations.

For the parameters in Table 2, the ordinary reference has poles $-159.414 \pm 5947.760i$ and $-21.836 \pm 1992.947i$, in s^{-1} . It therefore has two oscillatory modes. We fit stable scalar second- and third-order equations to its hammer angle, contact torque and hammer-side power on the complete contact interval. Each signal's residual is normalized by that reference signal's peak absolute value and weighted equally. The second-order fit has one complex pair; the third adds a real pole. Both retain initial angle and speed, and the third retains initial acceleration.

Table 1. Errors against the ordinary synthetic reference on the fitting interval.

Scalar order	Torque NRMSE	Hammer-power NRMSE
2, fitted	30.53%	25.67%
3, fitted	30.42%	25.61%
4, exact elimination	$< 10^{-10}\%$	$< 10^{-10}\%$

Fourth order retains the second oscillatory mode and recovers the torque and power pulse. Adding only a third derivative barely helps these fitted models. This establishes the order advantage for the stated reference; the unknown hammer hypothesis is evaluated by its own predictions below.

4 Signed work and apparent COP

The contact has two power ports. Define, over the complete contact interval $[0, T]$,

$$\begin{aligned} W_{hc} &= \int_0^T \tau_c \omega_h dt, & W_{ca} &= \int_0^T \tau_c \omega_a dt, \\ W_j &= \int_0^T \tau_j \omega_a dt, & W_X &= \int_0^T \tau_X \omega_h dt. \end{aligned} \quad (8)$$

Positive W_X means the unknown component receives net work; negative W_X means it supplies net work. Forward and returned contact work are both retained. This is the signed-account convention of Nilre and Herlin (2026b), Sections 2, 4 and 5.

For constant B_3, B_4 , write $v = \dot{\theta}_h$, $a = \ddot{\theta}_h$ and $j = \theta_h^{(3)}$. Integration by parts gives the exact identities

$$\begin{aligned} W_3 &= B_3[v a]_0^T - B_3 \int_0^T a^2 dt, \\ W_4 &= B_4[v j - \tfrac{1}{2} a^2]_0^T, & W_X &= W_3 + W_4. \end{aligned} \quad (9)$$

The third-order term includes an endpoint transfer and a signed integral; the fourth-order work is entirely an endpoint difference. Neither sign can be inferred from the coefficient alone. The fourth-order endpoint expression identifies the transfer without specifying a positive physical store or a replenishment mechanism.

Let $K_h = J_h \omega_h^2/2$, $K_a = J_a \omega_a^2/2$ and $U_j = k_j \theta_a^2/2$. For linear contact, $U_c = k_c \delta^2/2$ and $D_c = \int_0^T c_c \dot{\delta}^2 dt$. For (5), replace these by $U_c = 2H\delta^{5/2}/5$ and $D_c = \int_0^T H a \delta^{3/2} \dot{\delta}^2 dt$. In both cases $D_j = \int_0^T c_j \omega_a^2 dt$. Multiplying the component laws by their velocities gives

$$\begin{aligned} \Delta K_h &= W_d - W_{hc} - W_X, & W_d &= \int_0^T \tau_d \omega_h dt, \\ W_{hc} - W_{ca} &= \Delta U_c + D_c, & W_{ca} &= \Delta K_a + W_j, \\ W_j &= \Delta U_j + D_j. \end{aligned} \quad (10)$$

These balances follow directly from the declared torque laws.

The force-zero release rule can leave $U_c(T^-) > 0$. Removing that contact spring assigns $Q_{\text{rel}} = U_c(T^-)$ to a release account, with its physical destination unspecified. There is no body impulse at this event. The post-release ledger, with $K_0 = J_h \omega_h(0)^2/2$ and the stated zero initial anvil and elastic energies, is

$$K_0 + W_d - W_X = K_h(T) + K_a(T) + W_j + D_c + Q_{\text{rel}}. \quad (11)$$

Thus the endpoint contact store is counted once, as the release transfer. For the nonlinear zero-deformation release, $Q_{\text{rel}} = 0$.

Following the energy companion, exclude the unknown account from ordinary counted input. With $I = K_0 + W_d > 0$, report two apparent COPs:

$$\boxed{\text{COP}_a = \frac{W_{ca}}{I}, \quad \text{COP}_j = \frac{W_j}{I}.} \quad (12)$$

The first counts net work received by the anvil. The second counts work delivered to the modeled joint resistance and spring, including recoverable elastic storage. Their difference is $\Delta K_a/I$. In a real fastening

assessment, permanent tightening and the energy cost of preparing and replenishing the hammer would also have to be measured.

Anvil receipt and joint work answer different questions. The signed account also keeps work that returns during rebound from being counted as retained output.

5 Calculated impacts

All numerical values below are model calculations. The parameters are illustrative and common to the comparisons except for the explicitly stated contact law or higher-order weight.

Table 2. Common parameters.

Quantity	Value
Hammer inertia J_h	$1.6 \times 10^{-4} \text{ kg m}^2$
Anvil inertia J_a	$1.0 \times 10^{-4} \text{ kg m}^2$
Contact stiffness k_c	$1500 \text{ N m rad}^{-1}$
Contact damping c_c	$0.010 \text{ N m s rad}^{-1}$
Joint stiffness k_j	$1500 \text{ N m rad}^{-1}$
Joint damping c_j	$0.020 \text{ N m s rad}^{-1}$
Incoming hammer speed	600 rad s^{-1}
Counted input $I = K_0$	28.8 J

For the proposed models, $B_3 = 5.33333 \times 10^{-10} \text{ N m s}^3 \text{ rad}^{-1}$. The nominal fourth-order coefficient is $B_4 = 1.70667 \times 10^{-12} \text{ N m s}^4 \text{ rad}^{-1}$. The comparison with $w_4 = 1/5$ doubles this value and changes no other input. Table 3 gives pulse and outgoing-motion results; Figure 2 shows their time dependence.

Table 3. Complete-contact pulse results; outgoing speeds are in rad/s.

Model	T (ms)	Peak τ_c (N m)	$\omega_h(T)$	$\omega_a(T)$
Ordinary linear	1.5742	184.41	-560.17	-54.11
Ordinary nonlinear	1.3804	270.66	-522.92	-104.90
Third-order hammer	1.5740	185.33	-565.45	-52.92
Fourth, $w_4 = 1/10$	1.4796	233.14	-775.44	6.15
Fourth, $w_4 = 1/5$	0.7979	218.31	-358.25	-55.69

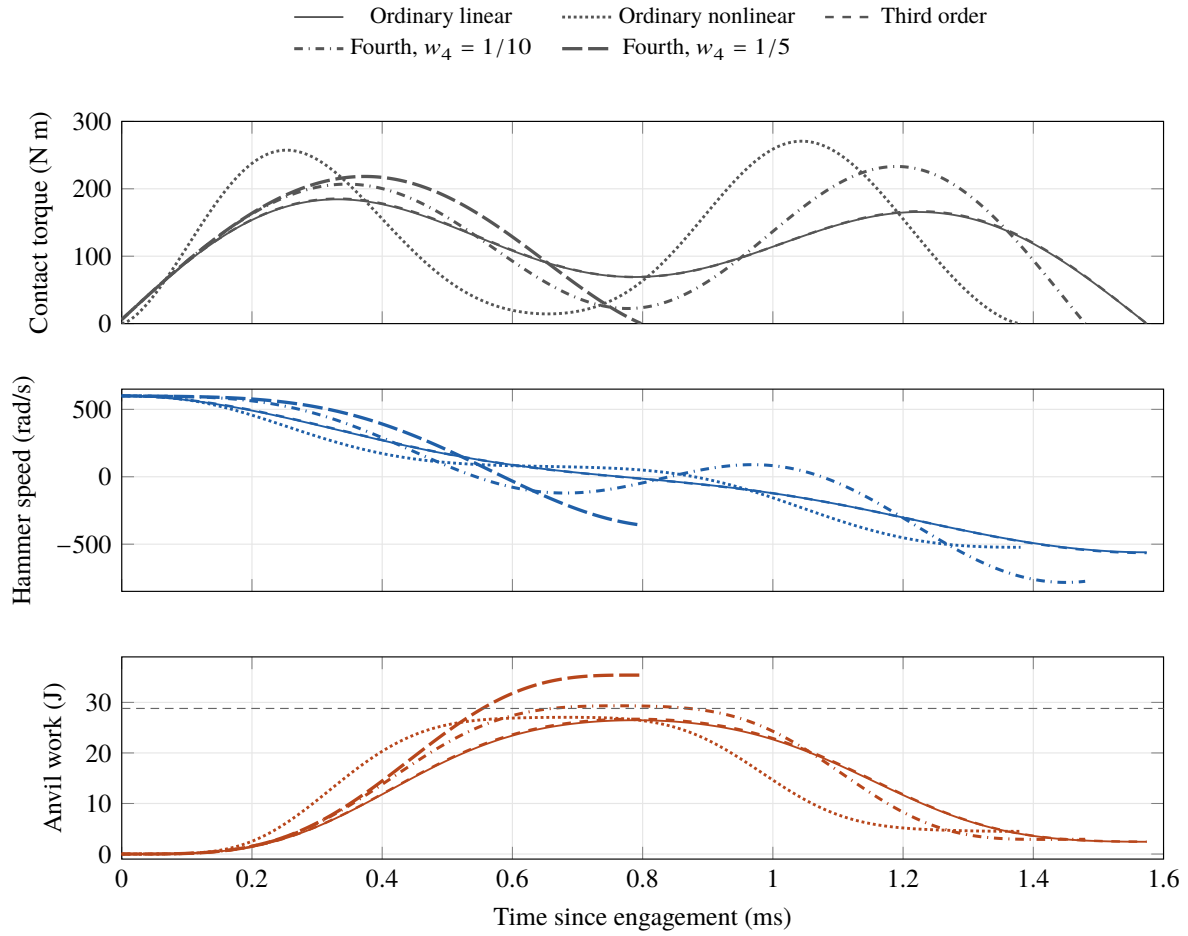


Figure 2. Calculated torque, hammer speed and cumulative anvil work for the common incoming state. Each curve stops at its own first separation. The horizontal work line is the counted input, 28.8 J. Returned work is retained in the falling portions of the lower curves.

Table 4. Signed interval work and apparent COP. Negative W_X is supplied work.

Model	W_{ca} (J)	W_j (J)	W_X (J)	COP_a	COP_j
Ordinary linear	2.4284	2.2820	0	0.08432	0.07924
Ordinary nonlinear	4.4938	3.9436	0	0.15603	0.13693
Third-order hammer	2.4487	2.3086	-0.5114	0.08502	0.08016
Fourth, $w_4 = 1/10$	2.9145	2.9126	-24.8868	0.10120	0.10113
Fourth, $w_4 = 1/5$	35.3945	35.2395	-18.1210	1.22898	1.22359

The ordinary linear contact sends 26.4708 J forward to the anvil and receives 24.0424 J back. Its net receipt is therefore only 2.4284 J. In the nominal fourth-order case, the corresponding amounts are 29.3476 and 26.4330 J. The unknown component supplies 24.8868 J, but the departing hammer retains 48.1048 J, compared with 25.1034 J in the ordinary case.

A large supplied-work contribution need not produce a large joint output. Here much of that work appears in the stronger hammer rebound.

For $w_4 = 1/5$, the unknown account supplies 18.1210 J, and both apparent COPs exceed one. The complete ledger is, to the displayed precision,

$$\underbrace{28.8000 + 18.1210}_{\text{ordinary input and supplied work}} = \underbrace{10.2673}_{K_h} + \underbrace{0.1550}_{K_a} + \underbrace{35.2395}_{W_j} + \underbrace{1.2561}_{D_c} + \underbrace{0.0031}_{Q_{rel}} \quad \text{J.} \quad (13)$$

Its joint work consists of 33.6402 J of elastic storage and 1.5993 J of resistance work. The higher-order contributions are $W_3 = -0.7819$ J and $W_4 = -17.3390$ J. Counting the supplied work as an additional input would give $W_{ca}/(I - W_X) = 0.7543$; the apparent ratio in (12) keeps the unknown account separate. The linear, nonlinear, third-order and nominal fourth-order release transfers are respectively 0.00854, 0, 0.00876 and 0.02036 J. Independent integrations and the endpoint identities in (9) check the accounts. Across the five examples, tighter calculations using a different integration method change COP by less than 10^{-10} and peak torque by less than 10^{-6} N m; the total balance residual is below 10^{-8} J.

6 Boundary, preparation and measurement

The calculations retain each model's coefficients when incoming speed is changed to 400 and 800 rad s^{-1} . For the linear-contact models, motion scales with incoming speed, work with its square, and apparent COP stays unchanged. The nonlinear contact gives anvil COPs of 0.12495 and 0.18297 at these speeds. These are predictions at changed conditions, without refitting.

The joint boundary matters even when the hammer coefficients are fixed. With $k_j = 1000$ N m rad^{-1} , the nominal fourth-order model separates at 0.6939 ms and gives $COP_a = 1.03358$, $COP_j = 0.87038$ and $W_X = -2.9530$ J. At $k_j = 2200$, its anvil COP is 0.07263. The changed boundary alters the return motion and which unloading zero ends the contact. The ordinary nonlinear model is also sensitive: its anvil COPs at those two stiffnesses are 0.92374 and 0.08413.

For fourth order, change only the prepared initial jerk to $\theta_h^{(3)}(0) = \pm \omega_h(0)/t_c^2$, where $t_c = \sqrt{J_h/k_c}$. The nominal model's anvil COP becomes 0.07467 for the negative choice and 0.13741 for the positive choice, compared with 0.10120 at zero jerk. The extra initial condition has an observable consequence and must be identified alongside the weights.

The nominal fourth-order engaged linear system has all poles in the left half-plane, with largest real part -17.51 s^{-1} . The softer-joint case above also has decaying modes. Doubling w_4 to $1/5$ introduces a

growing mode with real part 1095.19 s^{-1} . Its calculated blow ends after 0.7979 ms, before any assumption about later continuation is made. That growth is part of this parameter choice's prediction. Changing only w_4 to 1/20 instead gives anvil COP 0.08611 and decaying engaged modes.

Synchronized contact torque and hammer/anvil angular velocities would test these pulse and work predictions. The hammer residual $\tau_X = \tau_d - J_h \ddot{\theta}_h - \tau_c$ gives its inferred torque; integrating its product with ω_h tests the unknown account. Joint torque and anvil speed distinguish anvil receipt from joint work. Initial acceleration and jerk, separation timing, and independent changes of speed and joint stiffness are needed to distinguish the proposed weights from an unobserved preparation state. Derivatives should be inferred jointly from a motion model over a measured bandwidth; repeated differentiation of noisy samples can otherwise dominate the inferred higher-order torque.

These calculations resolve roughly 0.6–1.7 ms contacts over the stated finite parameter tests. Hardware accuracy and bandwidth remain to be established by those measurements. The useful comparison is whether one identified coefficient set predicts the pulse, rebound and signed work at independent impact conditions.

7 Conclusion

Synthesized third- and fourth-order coefficients give a concrete model of an unknown hammer contribution during one rotary impact. The ordinary two-inertia reference also explains why a fourth-order scalar description can outperform second- and third-order reductions: it retains both oscillatory modes.

The unknown component has a separate, computable work account. In the examples, its supplied work can strengthen hammer rebound, increase anvil receipt, or produce apparent COP above one. The outcome depends on the joint boundary, weights and prepared higher-order state. Signed contact and joint integrals, endpoint stores and release transfers identify where the ordinary accounted work goes. Measuring those ports and the hammer residual would test the predicted transfer and determine what physical account must accompany it.

References

1. H. Nilre and B. C. Herlin (2026a). *ODE Coefficient Synthesis: The coefficient lattice and its staircase*.
2. H. Nilre and B. C. Herlin (2026b). *Energy Ledgers for Forced Harmonic ODEs: Kirchhoff power balance and an unknown component with $X = LRC$* .
3. T. ter Braack and D. L. Margolis (2026). “Modeling of an Impact Wrench for Use in Reducing Hand–Arm Vibrations”. *Machines* **14**(2), 213. doi:10.3390/machines14020213.
4. K. H. Hunt and F. R. E. Crossley (1975). “Coefficient of Restitution Interpreted as Damping in Vibroimpact”. *Journal of Applied Mechanics* **42**(2), 440–445. doi:10.1115/1.3423596.